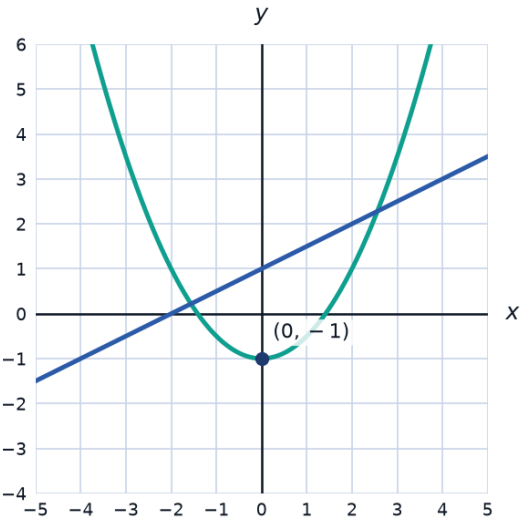
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains 8 red marbles, 6 green marbles, and 10 blue marbles. If 1 marble is chosen at random from the bag, what is the probability that the marble is NOT red?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, the line segment with endpoints $A(1, -2)$ and $B(5, 6)$ is perpendicularly bisected by line ℓ. If the point $(9, k)$ lies on line ℓ, what is the value of k?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The function $f(x) = a\sin(bx) + c$ has a maximum value of 9, a minimum value of -1, and a period of $\frac{\pi}{2}$. If $a > 0$ and $b > 0$, what is the value of $a + b + c$?

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what is the distance between the points $(-1, 2)$ and $(3, 5)$?

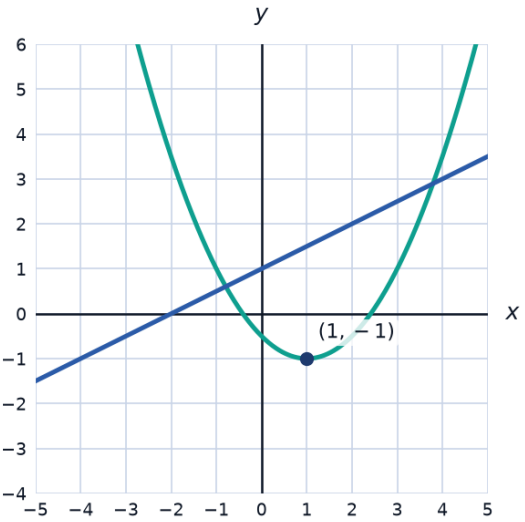
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k?

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = [2, -3; 1, 4]$ and $B = [-5, 1; 3, -2]$, what is the matrix $2A + B$?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints $(-3, 8)$ and $(5, -2)$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

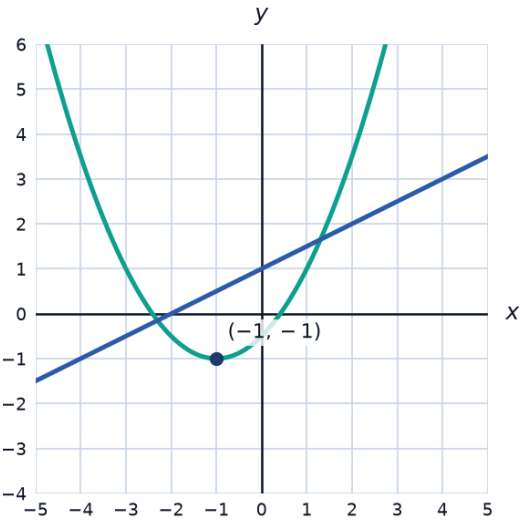
8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the standard (x, y) coordinate plane, what is the midpoint of the line segment with endpoints $(3, -4)$ and $(-5, 8)$?

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A fair coin is tossed twice. Find $P(\text{two heads})$.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A recipe is scaled from 2 to 5 servings. How many cups replace 4 cups?

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A square has area 36. Find its perimeter.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 3 and hypotenuse 5. Find $\sin\theta$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line passes through $(1, 2)$ with slope 3. Find y when $x = 4$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A random variable X has outcomes 7, 15, 19, 26 with $P(X = 7) = \frac{3}{12}, P(X = 15) = \frac{5}{12}, P(X = 19) = \frac{2}{12}, P(X = 26) = \frac{2}{12}$. Find $E(X)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

For $f(x) = x^2$, find the average rate of change on $[1, 3]$.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

If $\sin\theta = -\frac{9}{41}$ and θ is in Quadrant IV, find $\cos\theta$.

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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

A USD 80 item is marked down 20%. Find the sale price.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Practice 2.2: What is the sum of the solutions of the quadratic equation $3x^2 - 12x - 5 = 0$?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

Practice 2.1: For what values of x is $|2x - 3| < 7$?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

What is the sum and product of the roots of $3x^2 - 15x + 7 = 0$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the linear equation for x: $\frac{2x-3}{4} - \frac{x+1}{3} = \frac{5}{12}$.

a. 7

b. 8

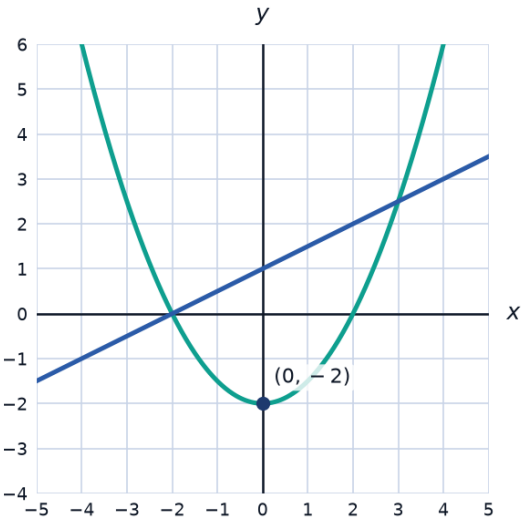
c. 9

d. 11

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k does the equation $4(2x - 5) + k = 8x - 13$ have infinitely many solutions?



a. 7

b. 13

c. 20

d. -7

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the complete solution set for the compound inequality $-7 \leq 3x + 2 < 11$?

a. $-3 \leq x < 3$

b. $-5 \leq x < 4$

c. $-3 < x \leq 3$

d. $x \geq -3$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A company sells widgets for 14 each with fixed costs of $1,800$ and variable costs of 5 per widget. How many widgets must be sold to make a profit of at least 900?

a. 200

b. 250

c. 300

d. 350

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the solutions to the quadratic equation $2x^2 - 7x - 15 = 0$?

a. $x = 5$ and $x = -\frac{3}{2}$

b. $x = -5$ and $x = \frac{3}{2}$

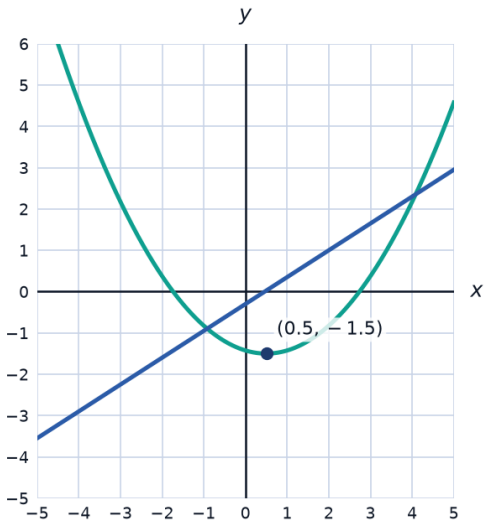
c. $x = 3$ and $x = -\frac{5}{2}$

d. $x = 5$ and $x = 3$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the sum of the solutions of the equation $3x^2 - 12x + 5 = 0$?



a. -4

b. 4

c. $\frac{5}{3}$

d. 12

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c will the quadratic equation $x^2 - 10x + c = 0$ have exactly one distinct real solution?

a. 5

b. 10

c. 25

d. 100

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the rational expression: $\frac{x^2 - 9}{2x^2 + 5x - 3}$ for all $x \neq \frac{1}{2}, -3$.

a. $\frac{x - 3}{2x - 1}$

b. $\frac{x + 3}{2x + 1}$

c. $\frac{x - 3}{2x + 1}$

d. $\frac{1}{2x - 1}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the radical equation for all valid real values: $\sqrt{3x + 13} = x + 3$.

a. $x = 1$ only

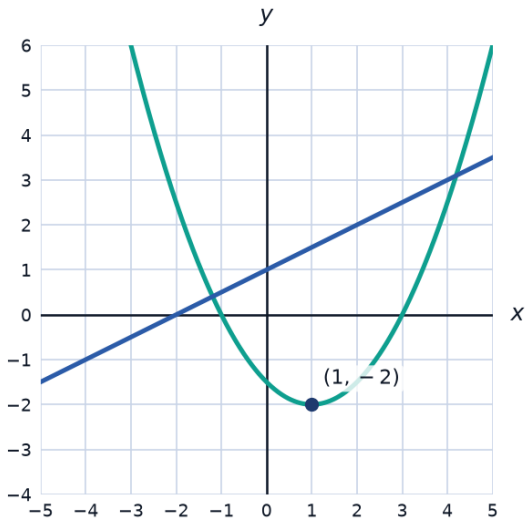
b. $x = -4$ only

c. $x = 1$ and $x = -4$

d. No real solutions

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 + 13x - 6 = 0$. What is the sum of the roots?



a. -17

b. -10

c. -16

d. -13

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the real numbers x and y satisfy the equation $2x^2 + y^2 - 4x + 6y + 11 = 0$ what is the value of $x + y$?

a. -4

b. -2

c. 2

d. 4

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following expressions is equivalent to $(5x^2 - 2x + 7) - (3x^2 - 6x - 4)$?

a. $2x^2 - 8x + 3$

b. $2x^2 - 8x + 11$

c. $2x^2 + 4x + 3$

d. $2x^2 + 4x + 11$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For all real values of x, which of the following expressions is equivalent to $(3x - 4)^2$?

a. $9x^2 - 24x + 16$

b. $9x^2 - 12x + 16$

c. $9x^2 - 16$

d. $9x^2 + 16$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve: $|x - 1| = 18$. What is the larger solution?

a. 13

b. 19

c. 24

d. 22

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve: $x^2 - 4x + 3 = 0$. What is the larger root?

a. 1

b. 5

c. 3

d. 0

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve: $|x| = 10$. What is the positive solution?

a. 9

b. 12

c. 10

d. 13

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve: $|x - 10| = 7$. What is the sum of all solutions?

a. 20

b. 16

c. 24

d. 25

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Evaluate $7^2 - 5^2$ using the difference of squares.

a. 19

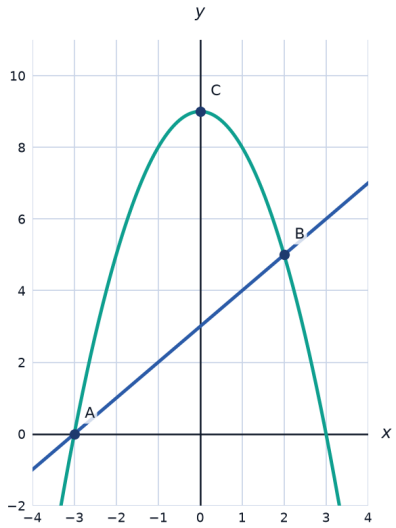
b. 25

c. 29

d. 24

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of ΔABC?



a. 12.5

b. 15

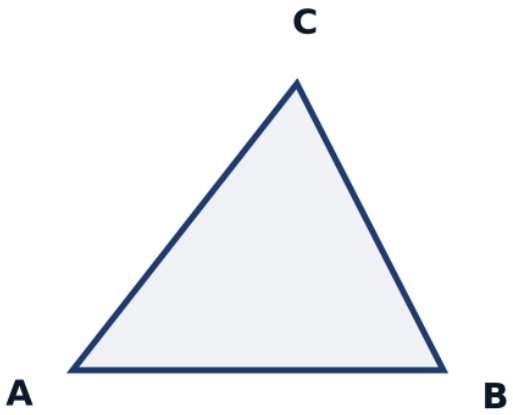
c. 22.5

d. 30

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40 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers**

In ΔABC shown below, the measure of $\angle A$ is 52° and the measure of $\angle B$ is 63°. What is the measure of $\angle C$?



a. 55°

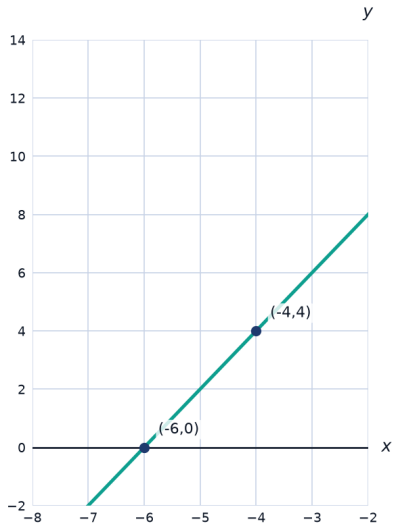
b. 65°

c. 75°

d. 115°

41 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios**

Find the slope of the line through $(-6, 0)$ and $(-4, 4)$.



a. $\frac{1}{5}$

b. 2

c. $\frac{18}{25}$

d. $\frac{2}{5}$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Functions & Graphs

A test consists of 25 multiple-choice questions. A student scored 80%. If the student correctly answered 4 more questions, what would the new percentage score be?

a. 84%

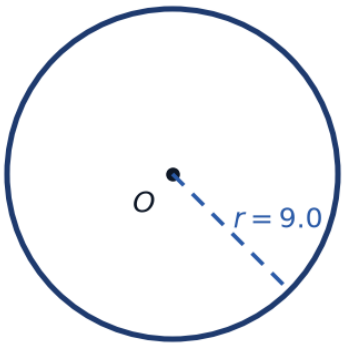
b. 88%

c. 92%

d. 96%

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Coordinate Geometry

A circle has radius 9. What is the arc length of a 180° sector (in terms of π)?



a. 8π

b. 9π

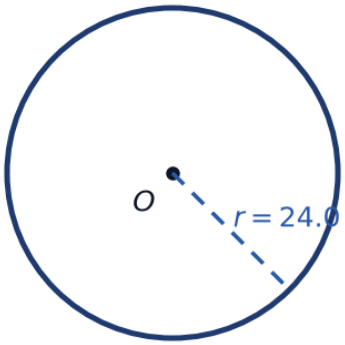
c. 11π

d. 18π

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

A circle has diameter 24. What is its circumference (in terms of π)?



a. 24π

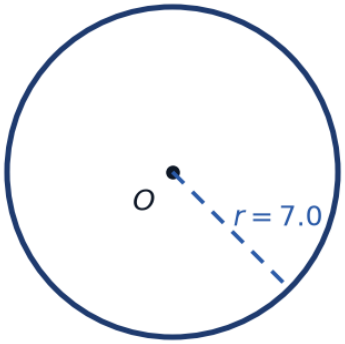
b. 144π

c. 48π

d. 12π

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

A circle has radius 7. What is its circumference (in terms of π)?



a. 28π

b. 14π

c. 7π

d. 49π

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